BRAC UNIVERSITY
Department of Computer Science and Engineering

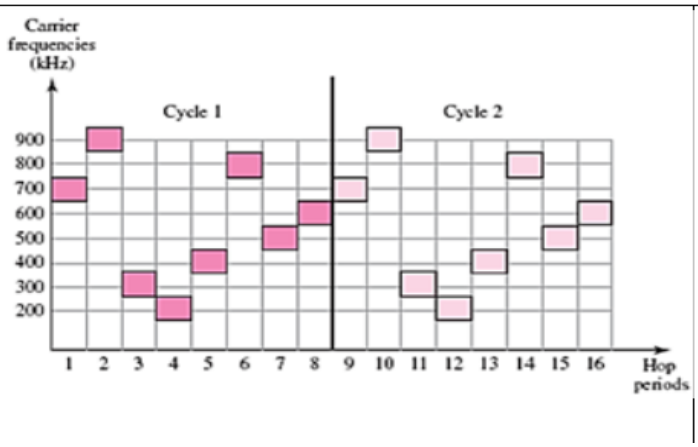
Examination: Semester Final | Semester: Spring 2024
Duration: 1 hr. 50 min | Full Marks: 50

CSE 320/EEE361/ECE361: Data Communications

Answer the following questions.
Figures in the right margin indicate marks.

SET A

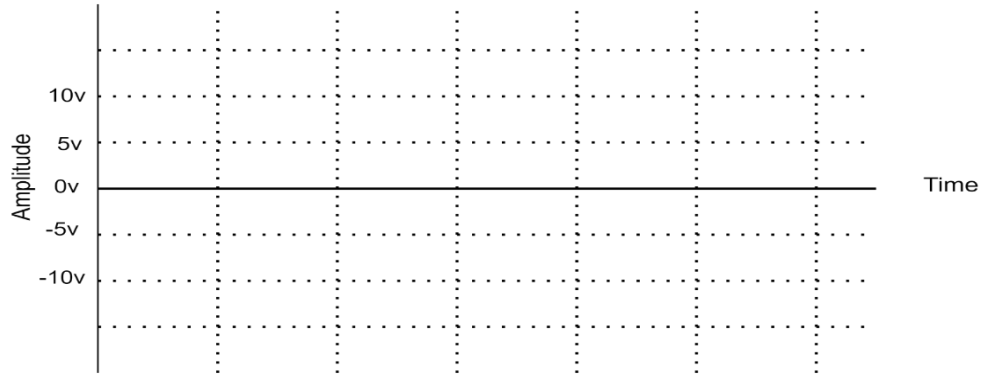
Name:	ID:	Section:

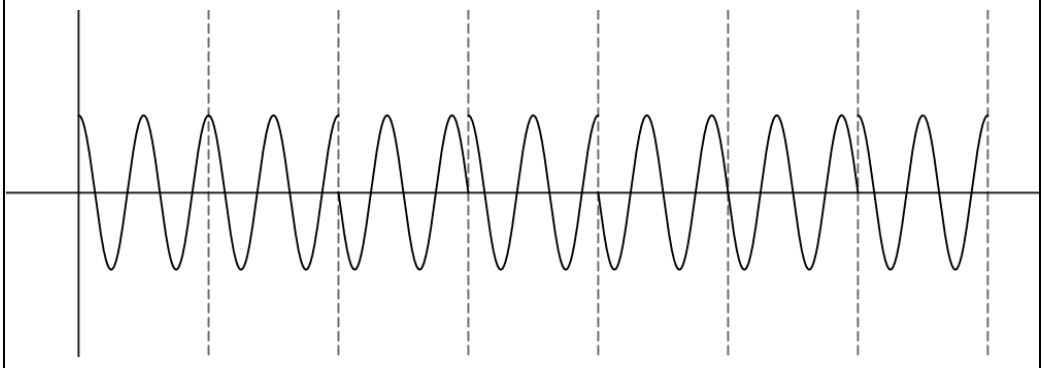
1. [CO3]	a)	Suppose 5 data sources with data rates of 240 Kbps, 120 Kbps, 120 Kbps, 240 Kbps and 480 Kbps are to be multiplexed with one synchronization bit. **Show** the visual representation of MUX with arbitrary input and output bit without adding extra pulses in the input channel.	[2]
[CO3]	b)	Suppose, each of the 5 channels send 6000 pages per minute where each page consists of 300 characters. If synchronous TDM is used to combine these sources using two-characters interleaving with one sync bit then answer the following questions: i. What is the input data rate (in bps) for each of the connections? ii. What is the input bit duration (in seconds)? iii. What is the frame rate (frames per second)? iv. What is the size of a frame in bits? v. What is the output data rate (in bps)? vi. What is the output bit duration (in seconds)?	[6]
[CO3]	c)	Suppose, the number of hopping frequencies in FHSS is 10, can we multiplex 10 channels into one by using the same B_{ss} bandwidth? Yes/no, explain briefly.  **Find** the pseudorandom generator's k-bit pattern output to match the given FHSS cycle and the Frequency hopping table.	[2+2]

k-bit	Frequency (kHz)
000	200
001	300
010	400
011	500
100	600
101	700
110	800
111	900

	d)	There is a digital medium with a data rate of 18 Mbps. How many 30-kbps voice channels can be carried by this medium if we use DSSS with the code sequence of '010110?	[2]
2. [CO2]	a)	For a noisy environment, which analog signal modulation is more suitable for signal transmission? Justify why.	[2]
[CO2]	b)	In a Multi-level FSK, for each signal element, we want to send 2 bits at a time. We have used a carrier signal that has an amplitude of 10v and phase is 0 degree. If the frequency changes according to the following table, **draw** the modulated signal for the bit sequence 1001000111. ***[answer this question in the given graph below]***	[5]
[CO2]	c)	**Determine** the digital bit stream from the analog signal below. The signal was modulated using Binary PSK where 0 means signal element with phase of π/2 rad and 1 means signal element with phase of π rad.	[5]

Bit Pattern	# cycles of the signal element
00	2
01	4
10	1
11	3





3. **[CO4]**	a)	Suppose two channels are using the following linear block coding scheme for error detection and correction. In a linear block coding scheme, XORing two distinct valid codewords produces another distinct codeword. Dataword | Codeword 00 | 00000 01 | 01011 10 | 10101 11 | ????? i. From the table above, find the value of n and k for the notation C(n,k). ii. What would be the missing distinct codeword? iii. Can we successfully detect and correct up to 2 bits error using the scheme?	[5]
[CO4]	b)	Assume the following cases for a sender and a receiver device that use CRC for error control. Dataword: $x^6 + x^4 + x^3 + x + 1$ Divisor: $x^4 + x^2 + x + 1$ **Show** the **sent codeword** in CRC **polynomial** format.	[5]
4. **[CO4]**	a)	A common multipoint link is being used by three stations. Station 1 is to the furthest left and Station 3 is to the furthest right, with the remaining stations in between. • Station 1 starts to send a frame to Station 3 at time t = 4 ms • The first bit of the frame reaches Stations 2 and 3 at times t = 13.4 ms and 17.1 ms respectively • Station 1 is supposed to send the last bit of the frame at time t = 21.4 ms i. Sketch a space and time representation of the channel if it is using CSMA/CD. ii. What is the Transmission Time and Maximum Propagation Time from the given information? iii. What should be the Minimum Frame Size and does it meet the requirements given above if the bandwidth is 2 Gbps?	[3+3+2]
[CO4]	b)	For a Pure ALOHA network, why is the Time-out waiting period twice the propagation time?	[2]
[CO4]	c)	For K=3, what is the size of the contention window in CSMA/CA? During the waiting time of the contention window slots, if the station finds the line busy, does the station halt or restart the contention window?	[2+2]